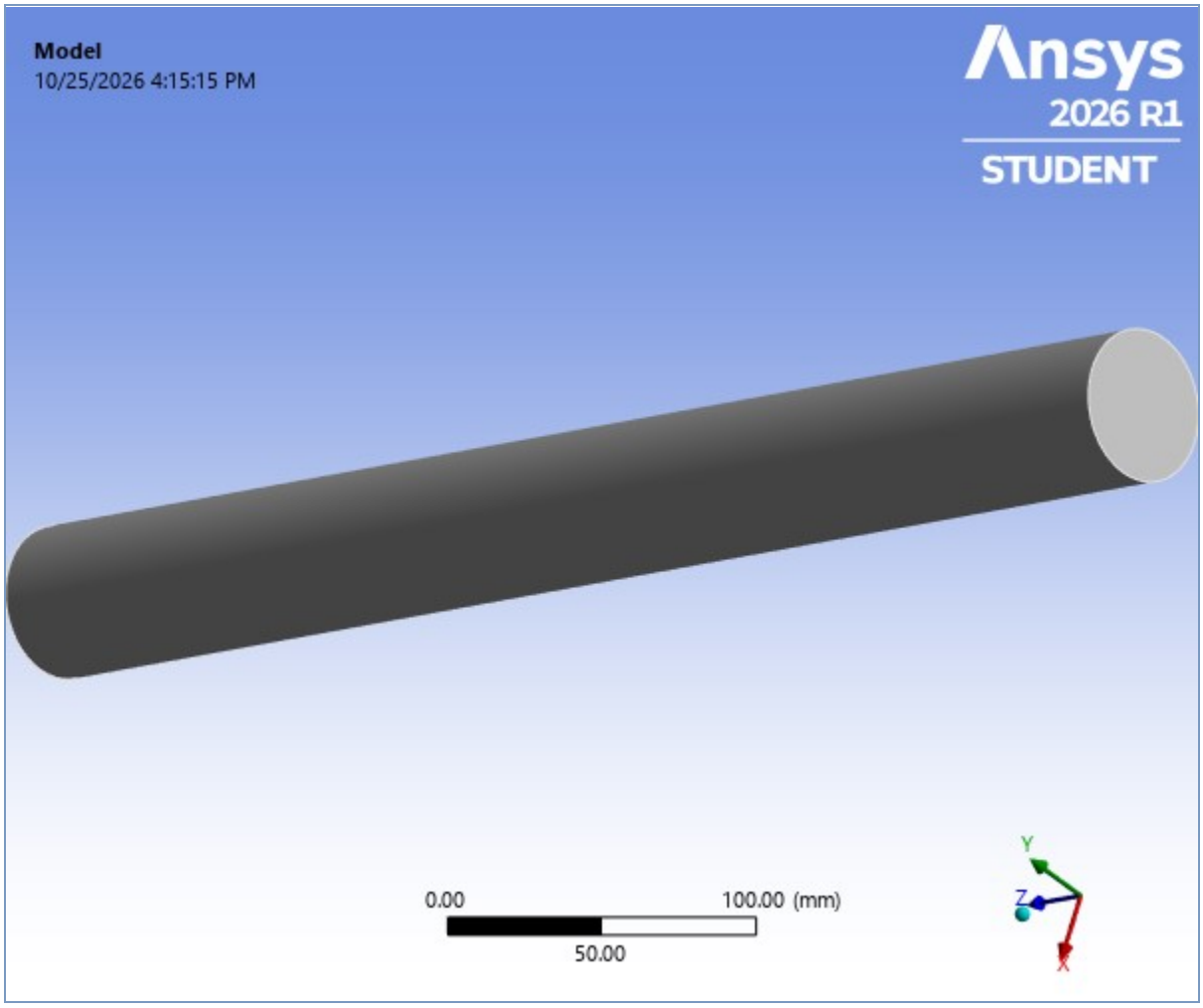
Project*

First Saved	Sunday, October 25, 2026
Last Saved	Sunday, October 25, 2026
Product Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4)**
  - Geometry Imports
    - Geometry Import (A3)
  - Geometry
    - SYS\Solid
  - Materials
  - Coordinate Systems
  - Mesh Workflows
  - Mesh
  - **Harmonic Acoustics (A5)**
    - Pre-Stress/Modal (None)
    - Analysis Settings
    - Acoustics Region
    - Loads
    - Solution (A6)
      - Solution Information
      - Results
- **Material Data**
  - Air

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	D:\Ansys_Temp\WB_User_6620_2\wbnew_files\dp0 \SYS\DM\SYS.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes

Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\Ansys_Temp\WB_User_6620_2\wbnew_files\dp0 \SYS\DM\SYS.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	50. mm
Length Y	50. mm
Length Z	500. mm
Properties	
Volume	9.8175e+005 mm³
Mass	1.2026e-003 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	44589
Elements	9792
Mesh Metric	None

Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	*SYS\Solid*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Air
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	50. mm
Length Y	50. mm
Length Z	500. mm
Properties	

Volume	9.8175e+005 mm³
Mass	1.2026e-003 kg
Centroid X	1.5386e-007 mm
Centroid Y	-1.8033e-017 mm
Centroid Z	250. mm
Moment of Inertia Ip1	25.16 kg·mm²
Moment of Inertia Ip2	25.16 kg·mm²
Moment of Inertia Ip3	0.37339 kg·mm²
Statistics	
Nodes	44589
Elements	9792
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	3
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

TABLE 8
Model (A4) > Mesh Workflows

Object Name	*Mesh Workflows*
State	Solved
Definition	
Active Mesh Workflow	

Mesh

TABLE 9
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	
Mesh Sizing	Adaptive
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	504.98 mm
Average Surface Area	27463 mm²
Minimum Edge Length	157.08 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0.1)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	44589

Elements	9792
Show Detailed Statistics	No

Harmonic Acoustics (A5)

TABLE 10
Model (A4) > Analysis

Object Name	*Harmonic Acoustics (A5)*
State	Solved
Physics Definition	
Structural	No
Acoustics	Yes
Definition	
Analysis Type	Harmonic Response
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4) > Harmonic Acoustics (A5) > Initial Condition

Object Name	*Pre-Stress/Modal (None)*
State	Fully Defined
Definition	
Modal Environment	None
Pre-Stress Environment	None Available

TABLE 12
Model (A4) > Harmonic Acoustics (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Multiple Steps	No
Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	1000. Hz
Solution Intervals	100
User Defined Frequencies	Off
Solution Method	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Scattering Controls	
Scattered Field Formulation	Program Controlled
Advanced	
Far-field Radiation Surface	Program Controlled
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	Yes

-- Value	Program Controlled
Calculate Velocity	No
Calculate Energy	No
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Ansys_Temp\WB_User_6620_2\wbnew_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 13
Model (A4) > Harmonic Acoustics (A5) > Acoustics Region

Object Name	*Acoustics Region*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Structural	No
Acoustics	Yes
Suppressed	No
Acoustic Domain Definition	
Artificially Matched Layers	Off
Advanced Settings	
Reference Pressure	2.e-011 MPa
Reference Static Pressure	0.10132 MPa
Fluid Behavior	Compressible

TABLE 14
Model (A4) > Harmonic Acoustics (A5) > Loads

Object Name	*Port*	*Port 2*	*Port In Duct*	*Port In Duct 2*	*Far-field Radiation Surface*
State	Fully Defined				
Port Surface(s)					
Scoping Method	Geometry Selection				
Geometry	1 Face				
Port Position	On Exterior Face				
Inside Surface Bodies					
Scoping Method	Geometry Selection				Geometry Selection
Geometry	1 Body				1 Body
Definition					
Type	Port		Port In Duct		Far-field Radiation Surface
Port Behavior	Transparent				
Suppressed	No				
Port Attribution			Inlet	Outlet	

Wave Type		Planar Wave		
Coordinate System		Global Coordinate System		
Pressure Amplitude		5.e-005 MPa		
Phase Angle		0. °		
Angle Phi (From X Axis Toward Y Axis)		0. °		
Angle Theta (From Z Axis Toward X-Y Plane)		0. °		
Port Selection		Port	Port 2	
Far-field Radiation Surface Selection				
Scoping Method				Geometry Selection
Geometry				1 Face

Solution (A6)

TABLE 15
Model (A4) > Harmonic Acoustics (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	4 m 38 s
MAPDL Memory Used	916. MB
MAPDL Result File Size	351.44 MB
Post Processing	
Beam Section Results	No

FIGURE 1
Model (A4) > Harmonic Acoustics (A5) > Solution (A6)

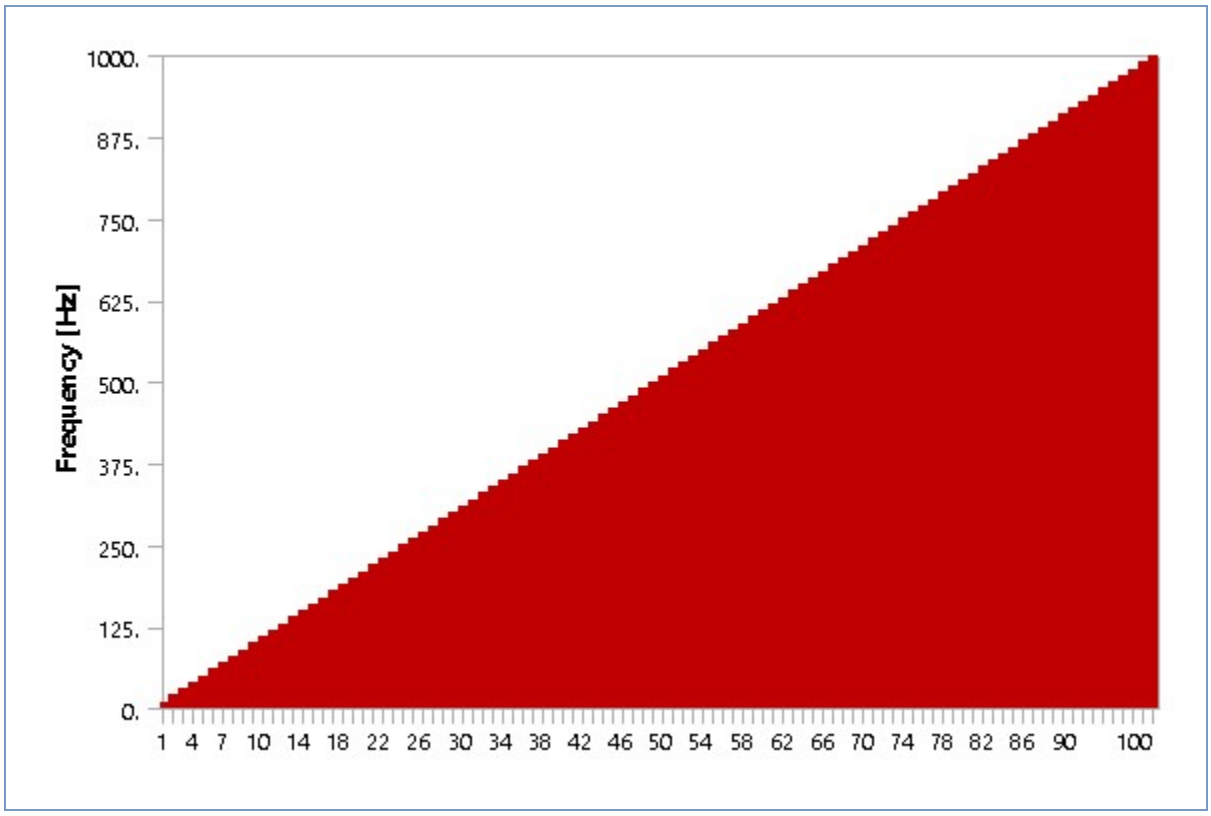


TABLE 16
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Results

Object Name	*Transmission Loss*	*Far-field SPL Mic*	*Far-field A-Weighted SPL Mic*
State	Solved		
Definition			
Type	Transmission Loss	Far-field SPL Mic	Far-field A-Weighted SPL Mic
Input Port	Port		
Output Port	Port 2		
Suppressed	No		
Definition Method		Coordinates	
Coordinate System		Global Coordinate System	

X Coordinate		0. mm	
Y Coordinate		0. mm	
Z Coordinate		-100. mm	
Reference RMS Sound Pressure		2.e-011 MPa	
Options			
Frequency Range	Use Parent		
Minimum Frequency	0. Hz		
Maximum Frequency	1000. Hz		
Results			
Minimum	0. dB	88.435 dB	18.005 dBA
Maximum	0. dB	100.13 dB	100.13 dBA
Advanced			
Model Type		3D	

FIGURE 2
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Transmission Loss

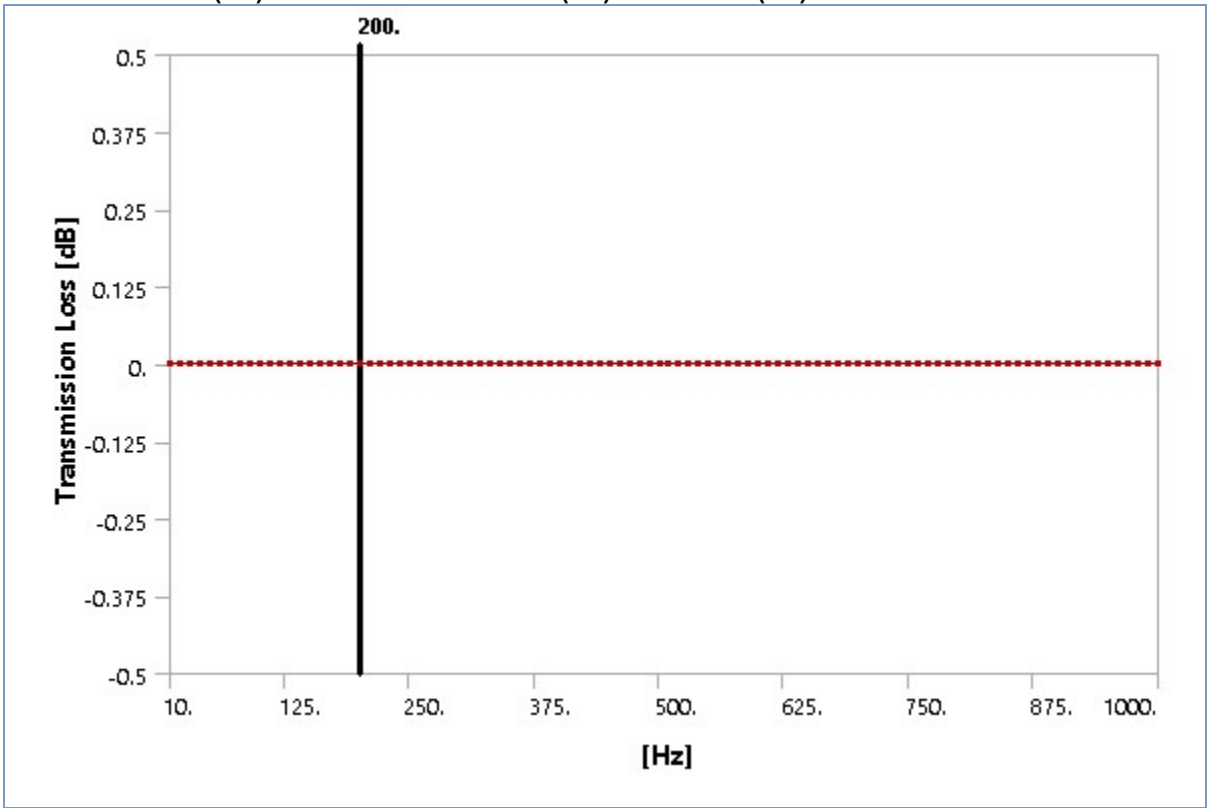


TABLE 18
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Transmission Loss

Frequency [Hz]	Transmission Loss [dB]
10.	
20.	
30.	
40.	
50.	
60.	
70.	
80.	
90.	

100.	
110.	
120.	
130.	
140.	
150.	
160.	
170.	
180.	
190.	
200.	
210.	
220.	
230.	
240.	
250.	
260.	
270.	
280.	
290.	
300.	
310.	
320.	
330.	
340.	
350.	
360.	0.
370.	
380.	
390.	
400.	
410.	
420.	
430.	
440.	
450.	
460.	
470.	
480.	
490.	
500.	
510.	
520.	
530.	
540.	
550.	
560.	
570.	
580.	
590.	
600.	
610.	
620.	

630.
640.
650.
660.
670.
680.
690.
700.
710.
720.
730.
740.
750.
760.
770.
780.
790.
800.
810.
820.
830.
840.
850.
860.
870.
880.
890.
900.
910.
920.
930.
940.
950.
960.
970.
980.
990.
1000.

FIGURE 3
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field SPL Mic

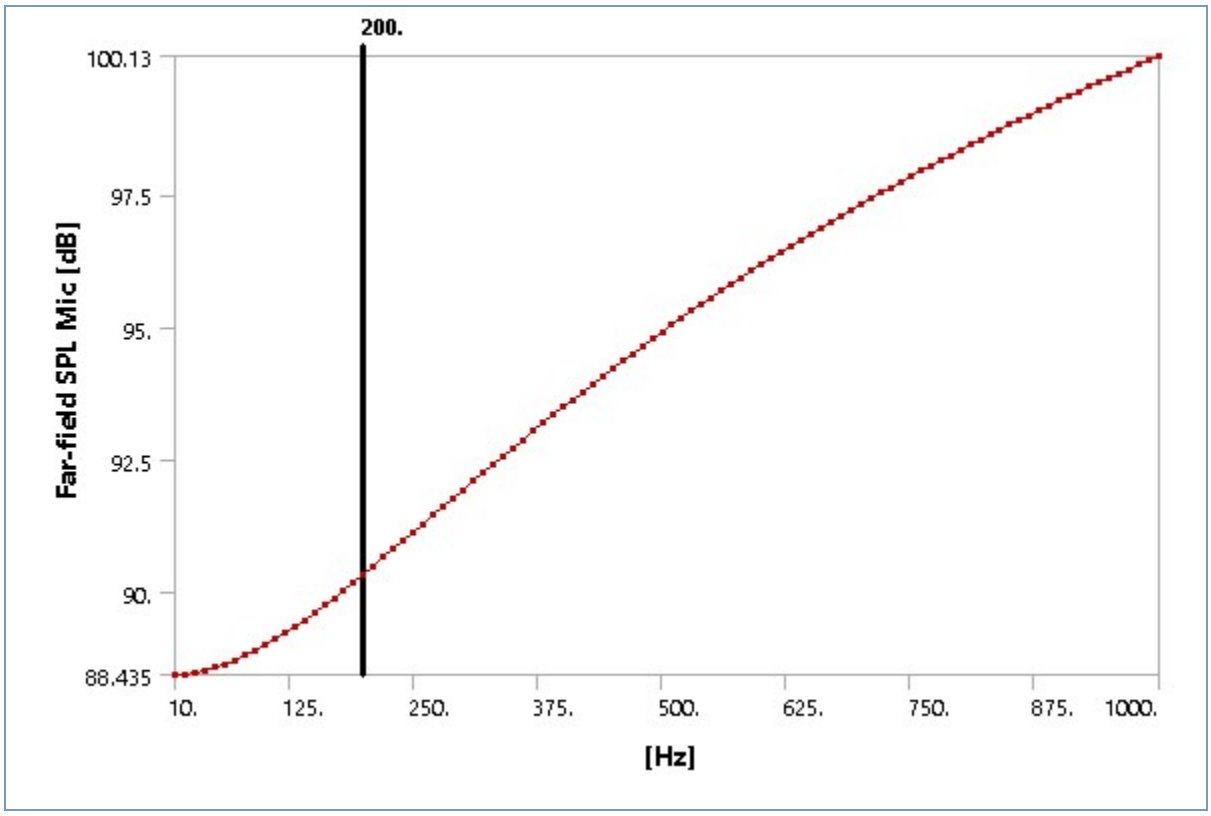


TABLE 19
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field SPL Mic

Frequency [Hz]	Far-field SPL Mic [dB]
10.	88.435
20.	88.453
30.	88.482
40.	88.524
50.	88.576
60.	88.639
70.	88.713
80.	88.796
90.	88.889
100.	88.99
110.	89.099
120.	89.215
130.	89.339
140.	89.468
150.	89.602
160.	89.742
170.	89.885
180.	90.033
190.	90.183
200.	90.336
210.	90.492
220.	90.649
230.	90.808
240.	90.967
250.	91.128

260.	91.289
270.	91.45
280.	91.611
290.	91.772
300.	91.933
310.	92.093
320.	92.252
330.	92.411
340.	92.568
350.	92.725
360.	92.88
370.	93.034
380.	93.187
390.	93.338
400.	93.488
410.	93.637
420.	93.784
430.	93.93
440.	94.075
450.	94.217
460.	94.359
470.	94.499
480.	94.637
490.	94.774
500.	94.909
510.	95.043
520.	95.176
530.	95.307
540.	95.436
550.	95.564
560.	95.691
570.	95.816
580.	95.94
590.	96.063
600.	96.184
610.	96.304
620.	96.422
630.	96.54
640.	96.656
650.	96.77
660.	96.884
670.	96.996
680.	97.107
690.	97.217
700.	97.326
710.	97.433
720.	97.539
730.	97.645
740.	97.749
750.	97.852
760.	97.954
770.	98.055
780.	

	98.155
790.	98.254
800.	98.352
810.	98.449
820.	98.546
830.	98.641
840.	98.735
850.	98.828
860.	98.921
870.	99.012
880.	99.103
890.	99.193
900.	99.282
910.	99.37
920.	99.457
930.	99.544
940.	99.63
950.	99.715
960.	99.799
970.	99.883
980.	99.966
990.	100.05
1000.	100.13

FIGURE 4
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field A-Weighted SPL Mic

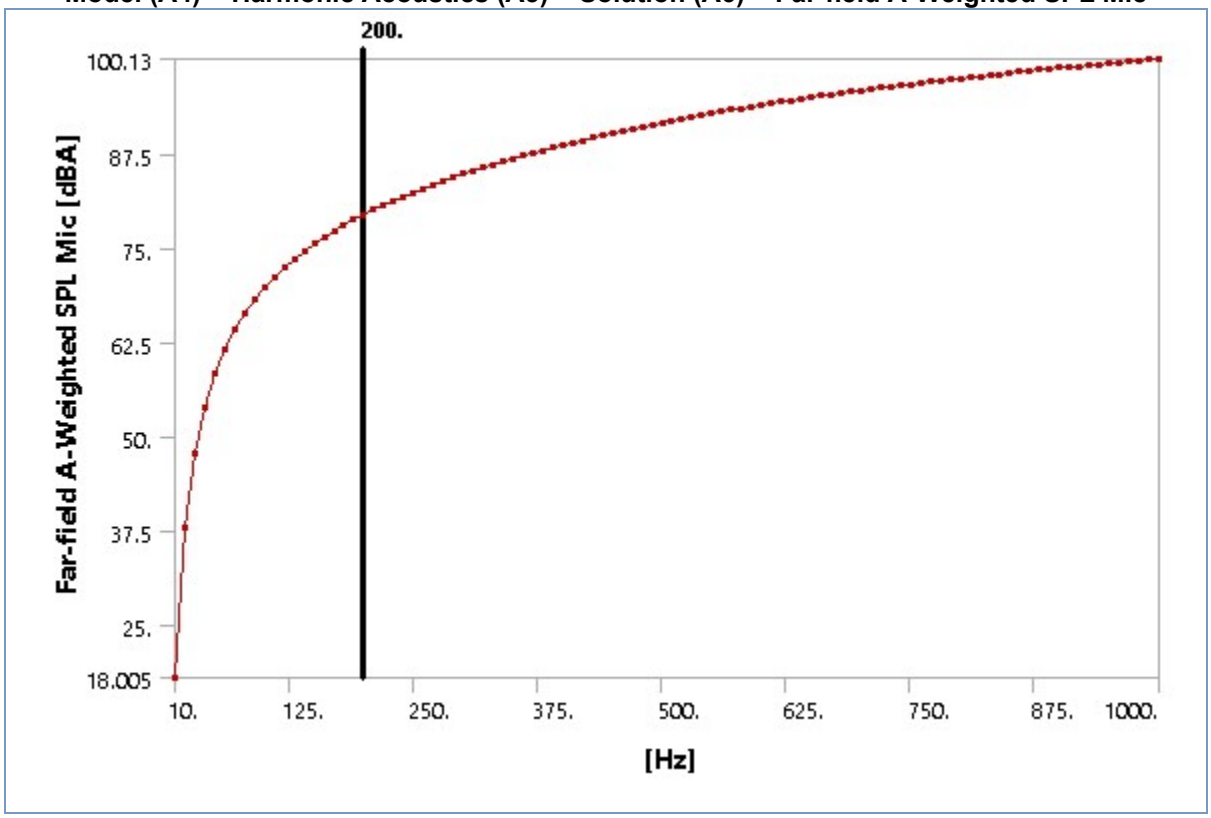


TABLE 20
Model (A4) > Harmonic Acoustics (A5) > Solution (A6) > Far-field A-Weighted SPL Mic

Frequency [Hz]	Far-field A-Weighted SPL Mic [dBA]
10.	18.005
20.	38.062
30.	47.88
40.	53.988
50.	58.304
60.	61.594
70.	64.224
80.	66.401
90.	68.248
100.	69.847
110.	71.254
120.	72.508
130.	73.64
140.	74.672
150.	75.62
160.	76.499
170.	77.317
180.	78.084
190.	78.807
200.	79.49
210.	80.139
220.	80.757
230.	81.347
240.	81.912
250.	82.454
260.	82.975
270.	83.476
280.	83.96
290.	84.427
300.	84.879
310.	85.316
320.	85.739
330.	86.15
340.	86.548
350.	86.935
360.	87.311
370.	87.676
380.	88.032
390.	88.378
400.	88.715
410.	89.043
420.	89.363
430.	89.675
440.	89.98
450.	90.277
460.	90.567
470.	90.85
480.	91.127
490.	91.397
500.	91.662
510.	91.92
520.	

	92.173
530.	92.421
540.	92.663
550.	92.9
560.	93.132
570.	93.359
580.	93.582
590.	93.8
600.	94.014
610.	94.224
620.	94.429
630.	94.631
640.	94.829
650.	95.023
660.	95.214
670.	95.401
680.	95.585
690.	95.765
700.	95.942
710.	96.117
720.	96.288
730.	96.456
740.	96.621
750.	96.784
760.	96.944
770.	97.101
780.	97.256
790.	97.408
800.	97.558
810.	97.705
820.	97.85
830.	97.993
840.	98.134
850.	98.272
860.	98.409
870.	98.543
880.	98.676
890.	98.807
900.	98.935
910.	99.062
920.	99.187
930.	99.311
940.	99.432
950.	99.552
960.	99.671
970.	99.788
980.	99.903
990.	100.02
1000.	100.13

Material Data

Air

TABLE 21
Air > Constants

Density	1.225e-009 kg mm^-3
Specific Heat	1.0064e+006 mJ kg^-1 K^-1
Thermal Conductivity	2.42e-005 W mm^-1 K^-1
Viscosity	1.7894e-011 MPa s

TABLE 22
Air > Molecular Weight

Molecular Weight kg kmol^-1
28.966

TABLE 23
Air > Lennard Jones Length

Lennard Jones Length mm
3711

TABLE 24
Air > Lennard Jones Energy

Lennard Jones Energy mJ
78600

TABLE 25
Air > Thermal Accom Coefficient

Thermal Accom Coefficient
0.9137

TABLE 26
Air > Velocity Accom Coefficient

Velocity Accom Coefficient mm s^-1
913.7

TABLE 27
Air > Formation Entropy

Formation Entropy mJ K^-1
1.9434e+008

TABLE 28
Air > Reference Temperature

Reference Temperature K
298.15

TABLE 29
Air > Critical Pressure

Critical Pressure mJ mm^-3
3.758

TABLE 30
Air > Critical Temperature

Critical Temperature K
132.3

TABLE 31
Air > Acentric Factor

Acentric Factor
3.3e-002

TABLE 32
Air > Critical Volume

Critical Volume mm^3
2.857e+006

TABLE 33
Air > Isotropic Relative Permeability

Relative Permeability
1

TABLE 34
Air > Absorption Coefficient

Absorption Coefficient mm^-1
0

TABLE 35
Air > Speed of Sound

Speed of Sound mm s^-1
3.4625e+005